



Lecture 11: Sum of a random number of random variables

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1. Review
2. Sum of independent RVs
3. Sum of a random number of independent RVs
4. Recitation
5. Summary

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- Conditional expected value

$$\mathbf{E}[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Law of iterated expectations

$$\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X|Y]] = \sum_y \mathbf{E}[X|Y = y] p_Y(y)$$

- Conditional variance

$$\text{var}(X|Y) = \mathbf{E}[(X - \mathbf{E}[X | Y])^2 | Y] = \mathbf{E}[\tilde{X}^2 | Y]$$

- Law of total variance

$$\text{var}(X) = \text{var}(\mathbf{E}[X|Y]) + \mathbf{E}[\text{var}(X|Y)]$$

- Transforms

$$M_X(s) = \mathbf{E}[e^{sX}]$$

- Sanity checks

$$M_X(0) = 1$$

$$|M_X(s)| \leq 1 \text{ for } \operatorname{Re}(s) = 0$$

- Linear operations

$$M_{aX+b}(s) = e^{sb} M_X(sa)$$

$$M_{X+Y}(s) = M_X(s)M_Y(s) \text{ if } X, Y \text{ independent}$$

- Expected values

$$\mathbf{E}[X^n] = \left. \frac{\partial^n M_X(s)}{\partial s^n} \right|_{s=0}$$

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Sum of independent RVs

Let X and Y be independent RVs, and let $Z = X + Y$. The Transform associated with Z is

$$M_Z(s) = M_X(s)M_Y(s) .$$

- **Derivations:** The Transform associated with Z is

$$M_Z(s) = \mathbf{E}[e^{sZ}] = \mathbf{E}[e^{s(X+Y)}] = \mathbf{E}[e^{sX}e^{sY}] .$$

Consider the independence of e^{sX} and e^{sY} ,

$$M_Z(s) = \mathbf{E}[e^{sX}]\mathbf{E}[e^{sY}] = M_X(s)M_Y(s) .$$

- **Generalization:** If $X_1, \dots, X_n$ is a collection of independent RVs, and $Z = X_1 + \dots + X_n$, then

$$M_Z(s) = M_{X_1}(s) \cdots M_{X_n}(s) .$$

- Let $X_1, \dots, X_n$ be independent **Bernoulli** RVs with a common parameter p . Then,

$$M_{X_i}(s) = (1 - p)e^{0s} + pe^{1s} = 1 - p + pe^s, \quad \text{for all } i.$$

- The RV $Z = X_1 + \dots + X_n$ is **binomial** with parameters n and p .
The corresponding Transform

$$M_Z(s) = (1 - p + pe^s)^n .$$

- Let X and Y be independent **Poisson** RVs with means λ and μ , and let $Z = X + Y$. Then, Z is still **Poisson**, with mean $\lambda + \mu$.

$$M_Z(s) = M_X(s)M_Y(s) = e^{\lambda(e^s-1)}e^{\mu(e^s-1)} = e^{(\lambda+\mu)(e^s-1)} .$$

- Let X and Y be independent **Gaussian** RVs with means μ_x, μ_y , and variances σ_x^2, σ_y^2 . Let $Z = X + Y$. Then, Z is still **Gaussian**, with mean $\mu_x + \mu_y$ and variance $\sigma_x^2 + \sigma_y^2$.

$$M_X(s) = \exp \left\{ \frac{\sigma_x^2 s^2}{2} + \mu_x s \right\}, \quad M_Y(s) = \exp \left\{ \frac{\sigma_y^2 s^2}{2} + \mu_y s \right\},$$

and

$$M_Z(s) = \exp \left\{ \frac{(\sigma_x^2 + \sigma_y^2) s^2}{2} + (\mu_x + \mu_y) s \right\}.$$

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- Consider the sum

$$Y = X_1 + \cdots + X_N ,$$

where N is a RV that takes integer values, and $X_1, X_2, \dots, X_N$ are identically distributed RVs.

- Assume that $N, X_1, X_2, \dots$ are **independent**.
- First condition on the event $\{N = n\}$, which yields the case of a fixed number of RVs.

$$\begin{aligned} \mathbf{E}[Y|N = n] &= \mathbf{E}[X_1 + X_2 + \cdots + X_n|N = n] \\ &= \mathbf{E}[X_1 + X_2 + \cdots + X_n] \\ &= \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n] \\ &= n \mathbf{E}[X] . \end{aligned}$$

- This is true for every nonnegative integer n , so

$$\mathbf{E}[Y \mid N] = N \mathbf{E}[X] .$$

- Using the law of iterated expectations, we obtain

$$\begin{aligned} \mathbf{E}[Y] &= \mathbf{E}[\mathbf{E}[Y \mid N]] \\ &= \mathbf{E}[N \mathbf{E}[X]] \\ &= \mathbf{E}[N] \mathbf{E}[X] . \end{aligned}$$

- Firstly,

$$\mathbf{E}[Y|N] = N \mathbf{E}[X]$$

$$\text{var}(\mathbf{E}[Y|N]) = (\mathbf{E}[X])^2 \text{var}(N)$$

- Secondly,

$$\text{var}(Y|N = n) = n \text{var}(X)$$

$$\text{var}(Y|N) = N \text{var}(X)$$

$$\mathbf{E}[\text{var}(Y|N)] = \mathbf{E}[N] \text{var}(X)$$

- Using the law of total variance to obtain

$$\begin{aligned} \text{var}(Y) &= \mathbf{E}[\text{var}(Y|N)] + \text{var}(\mathbf{E}[Y|N]) \\ &= \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N) . \end{aligned}$$

- Conditioned on $N = n$, Y is the sum of the independent RVs $X_1, \dots, X_n$,

$$\begin{aligned}\mathbf{E}[e^{sY} | N = n] &= \mathbf{E}[e^{sX_1} \dots e^{sX_n} | N = n] \\ &= \mathbf{E}[e^{sX_1} \dots e^{sX_n}] \\ &= \mathbf{E}[e^{sX_1}] \dots \mathbf{E}[e^{sX_n}] \\ &= (M_X(s))^n .\end{aligned}$$

- Using the law of iterated expectations,

$$\begin{aligned}M_Y(s) &= \mathbf{E}[e^{sY}] = \mathbf{E}[\mathbf{E}[e^{sY} | N]] \\ &= \mathbf{E}[(M_X(s))^N] \\ &= \sum_{n=0}^{\infty} (M_X(s))^n p_N(n) .\end{aligned}$$

- Using the observation

$$(M_X(s))^n = e^{\log(M_X(s))^n} = e^{n \log M_X(s)} .$$

- We obtain

$$M_Y(s) = \sum_{n=0}^{\infty} e^{n \log M_X(s)} p_N(n) .$$

- Comparing with the formula

$$M_N(s) = \mathbf{E}[e^{sN}] = \sum_{n=0}^{\infty} e^{sn} p_N(n) .$$

– $M_Y(s)$ is obtained from $M_N(s)$ with s replaced by $\log M_X(s)$.

Summary on Properties

Consider the sum

$$Y = X_1 + \cdots + X_N ,$$

where N is a RV that takes integer values, and $X_1, X_2, \dots$ are identically distributed RVs. Assume that $N, X_1, X_2, \dots$ are independent.

- Mean

$$\mathbf{E}[Y] = \mathbf{E}[N]\mathbf{E}[X]$$

- Variance

$$\text{var}(Y) = \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N)$$

- Transform

$$M_Y(s) = M_N(\log M_X(s))$$

- Assume that N and X_i are both geometrically distributed with parameter p, q , respectively.
- All of these RVs are independent. Define $Y = X_1 + \cdots + X_N$.
- We have

$$M_N(s) = \frac{pe^s}{1 - (1-p)e^s}, \quad M_X(s) = \frac{qe^s}{1 - (1-q)e^s}.$$

- Replacing each occurrence of e^s with $M_X(s)$ yields

$$M_Y(s) = \frac{pM_X(s)}{1 - (1-p)M_X(s)} = \frac{pqe^s}{1 - (1-pq)e^s}.$$

- Y is also geometrically distributed, with parameter pq .

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1. You are visiting the rainforest, but unfortunately your insect repellent has run out. As a result, at each second, a mosquito lands on your neck with probability 0.5. If one lands, with probability 0.2 it bites you, and with probability 0.8 it never bothers you, independently of other mosquitoes.

- (a) What is the expected time between successive mosquito bites? What is the variance of the time between successive mosquito bites?
- (b) In addition, a tick lands on your neck with probability 0.1. If one lands, with probability 0.7 it bites you, and with probability 0.3, it never bothers you, independently of other ticks and mosquitoes. Now, what is expected time between successive bug bites? What is the variance of the time between successive bug bites?

2. Suppose there are n papers in a drawer. You draw a paper and sign it, and then, instead of filing it away, you place the paper back into the drawer. If any paper is equally likely to be drawn each time, independent of all other draws, what is the expected number of papers that you will draw before signing all n papers?

3. Show that the sum of a Poisson-distributed number of i.i.d. Bernoulli random variables is Poisson.

Hint: The transform of Poisson RV with parameter λ is $M_X(s) = e^{\lambda(e^s-1)}$, and the transform of Bernoulli RV with parameter p is $M_X(s) = 1 - p + pe^s$.

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- Sum of independent RVs

$$M_Z(s) = M_{X_1}(s) \cdots M_{X_n}(s)$$

- Sum of a random number of independent RVs

$$\mathbf{E}[Y] = \mathbf{E}[N] \mathbf{E}[X]$$

$$\text{var}(Y) = \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N)$$

$$M_Y(s) = \sum_{n=0}^{\infty} (M_X(s))^n p_N(n) = M_N(\log M_X(s))$$

- Readings: Section 4.4-4.5
- Assignment: Problem Set 11
- Next lecture: Weak law of large numbers